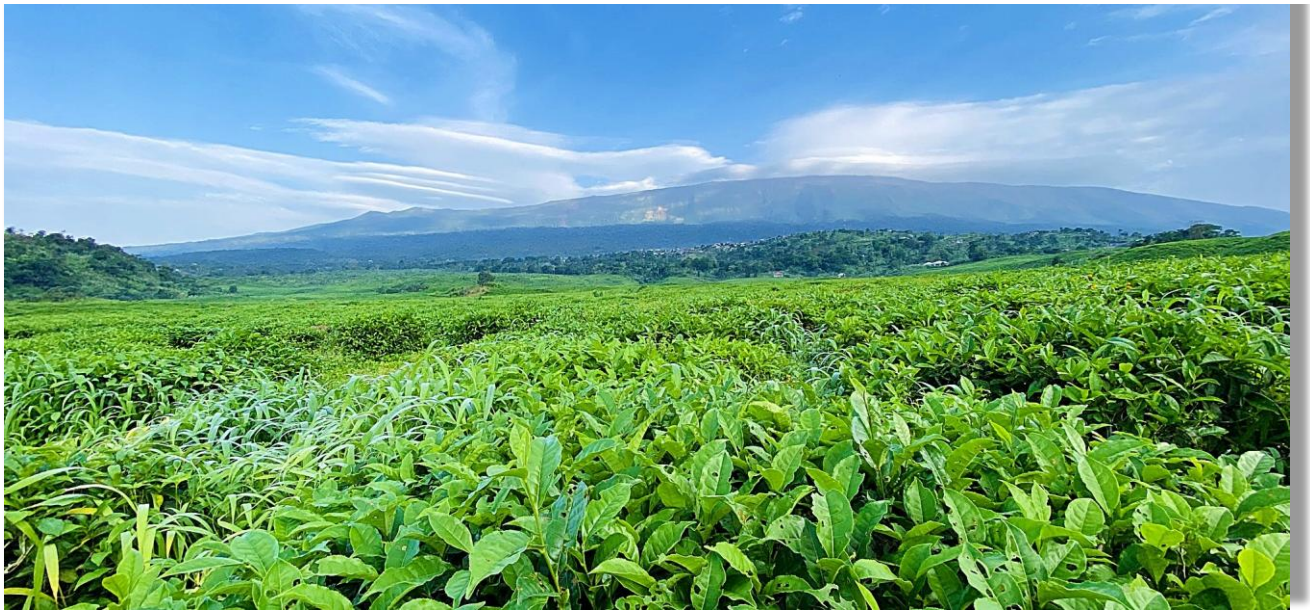



COUNTRY DOSSIER

# Cameroon

Water, climate, fertilisers and crops: National study and a study of each region, potential benefits of our products, external trade

Green Solutions · contact@green-solutions-group.com · green-solutions-group.com



*The Tole tea estate, below Mount Cameroon (South-West).*

# Foreword

This dossier describes Cameroon from the angle of water, climate, fertilisers and crops. It is written for the ministries in charge of agriculture, water and trade, the regional directorates of agriculture, cooperatives and producer organisations. It answers three questions: Where is water short, for which crops, and in which regions? It then examines what our products can change, in rainfed as well as irrigated farming, and what that can mean for the country's external trade.

**Reading the sources.** Every figure refers to a source, named in the text and detailed in Annex A. Each source carries a confidence level:

**How to read this document**

- **A**: Official statistics or international database (World Bank, FAO, CEMAC, ECCAS, Cameroonian administration).
- **B**: Research institute, laboratory, United Nations agency or recognised climate model.
- **C**: Manufacturer statement, website, press or estimate.
- "M" means million; "Mm³" means million cubic metres; "t" means tonnes; "I$" means constant 2014-2016 international dollars.
- A "scenario" is a calculation based on the trials cited. It is not a measurement made in Cameroon.

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# Summary

The points below summarise the dossier. The figures come from public sources: The World Bank, FAO (FAOSTAT, AQUASTAT), CEMAC, ECCAS, Cameroonian administration, PVGIS and ERA5. The production gains and water savings are scenarios, described in Annex B.

**29.88 M**
Inhabitants in 2025 (World Bank)

**1.20 M ha**
Of maize (corn) harvested in 2024, the main crop

**28,700 ha**
Equipped for irrigation (2024)

**+1,568 t**
Of maize per 10,000 ha treated, central scenario

**160 Mm³**
Of irrigation water saved a year, central scenario

**2,059 M$**
Of agricultural exports in 2024 (FAOSTAT)

**Key points**

• **A rain-fed agriculture.** 28,700 ha equipped for irrigation in 2024, for 7.97 million hectares of cropland: Almost every harvest depends on rain.

• **Maize (corn): The main crop.** 1,200,000 ha harvested in 2024, for 2.20 million tonnes, or 1.83 t/ha.

• **Water is not short over the year, it is short between two rains.** National water stress is only 1.6%, but at Maroua, the air demands 1,671 mm more than it rains.

• **In rainfed farming, the gain is counted in tonnes.** In the central scenario, 1,568 t of maize, 991 t of soybean and 16,669 t of potato more per 10,000 ha treated.

• **In irrigation, the water saved enlarges the schemes.** 160.5 Mm³ a year in the central scenario, enough to irrigate about 6,249 more ha, or 21.8% of the equipped area.

• **An organic chain.** 2,820 ha of certified organic cropland in 2024: ECOFERT® is EU Bio certified.

• **All nitrogen is imported.** 52,002 t of nitrogen used in 2024, 52,526 t imported, no national production.

• **Farm trade.** $2,059 million of exports and $1,816 million of imports in 2024. First sale: Cocoa beans ($1,170 million). Rice weighs $545 million of purchases.

## Key figures

### Key indicators for Cameroon

| Indicator | Value | Year | Source (level) |
|---|---|---|---|
| Population | 29.88 M | 2025 | World Bank (A1) |
| GDP per capita | 1,972 $ | 2025 | World Bank (A1) |
| Agriculture share of GDP | 16.8% | 2025 | World Bank (A1) |
| Agricultural employment | 41.9% | 2025 | World Bank (A1) |
| Cropland | 7,967,020 ha | 2024 | FAOSTAT (A6) |
| Area equipped for irrigation | 28,700 ha | 2024 | FAOSTAT (A6) |
| Harvested area: Maize (corn) | 1,200,000 ha | 2024 | FAOSTAT (A2) |
| Certified organic cropland | 2,820 ha | 2024 | FAOSTAT (A6) |
| Agricultural exports | 2,059 M$ | 2024 | FAOSTAT (A4) |
| Agricultural imports | 1,816 M$ | 2024 | FAOSTAT (A4) |

| Indicator | Value | Year | Source (level) |
|---|---|---|---|
| Mineral nitrogen, agricultural use | 52,002 t | 2024 | FAOSTAT (A5) |

*Source: Table based on the sources indicated.*

# 1. Green Solutions and our products

Green Solutions develops products for water, fertilisation and crop protection. Four families are of interest to this dossier. The yields cited are those published by the manufacturer or by the trials, with their confidence level.

### Products covered by this dossier

| Product | Claimed use | Published evidence (level) |
|---|---|---|
| EVERGREEN® (water-retention polymer) | Holds rain or irrigation water as a gel, near the roots. | Trials on maize, potato, watermelon, date palm and soybean. Level B to C. |
| ECOSORB® (water-retention polymer based on corn starch) | Same use; action lasting 2 to 3 years according to the website. | Soybean trial (China); laboratory acrylamide analysis. Level B to C. |
| ECOFERT® (fertiliser) | Fertiliser certified by SOHISCERT and EU Bio, usable in organic farming. | Published certificate; potato trial combined with EVERGREEN®. Level A (certificate), C (results). |
| Paulownia ALTIFOLIA® | Fast-growing tree for agroforestry. | Website figures; no trial in Cameroon. Level C. |
| Water Vital®, NAPEMA® | Outside the scope of this dossier. | No independent source in the documents consulted. |

*Source: Green Solutions website (product sheets and published trials), Annex A.*

The indicative dose published on the website is 30 to 75 kg/ha.



*A rainy-season sorghum field at Kousseri, in the Far North.*

# 2. Cameroon in figures

Cameroon had 29.88 million inhabitants in 2025, 44% of them in rural areas. Agriculture accounts for 16.8% of gross domestic product and employs 42% of the labour force. The country covers 472.7 thousand km², of which 21% is agricultural land. Undernourishment affects 4.9% of the population.

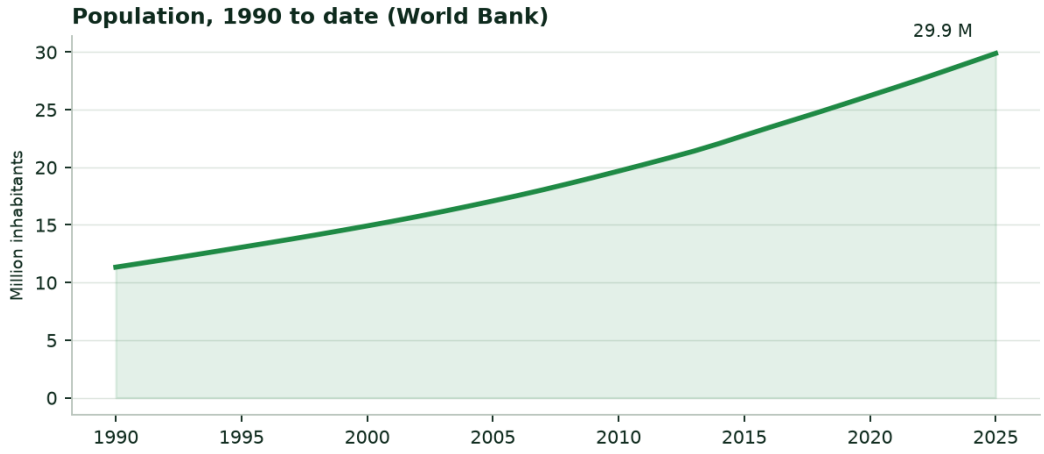



*Figure 1. Population of Cameroon, 1990 to date. Source: World Bank (A1)*

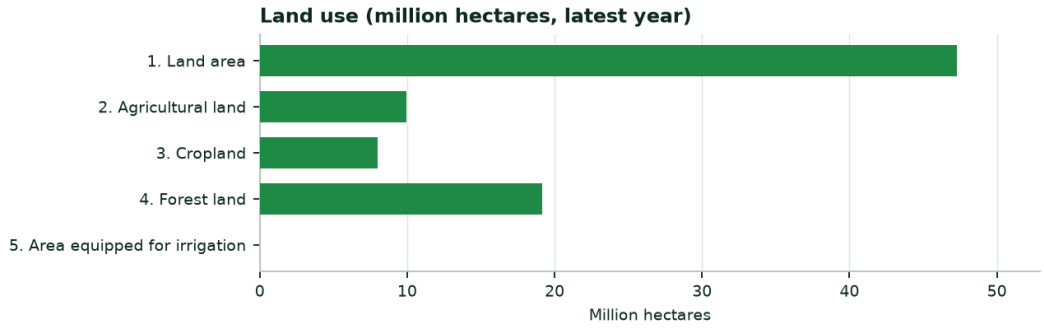



*Figure 2. Land use, latest year published. Source: FAOSTAT (A6)*

### Agricultural and water indicators, latest year published

| Indicator | Value | Year | Source |
|---|---|---|---|
| Cropland certified organic | 2,820 ha | 2024 | FAOSTAT (A6) |
| Average cereal yield | 1,770 kg/ha | 2024 | World Bank (A1) |
| Fertiliser use per hectare of arable land | 16.3 kg/ha | 2024 | World Bank (A1) |
| Water withdrawals, agricultural share | 67.7% | 2022 | World Bank (A1) |
| Internal freshwater resources | 273.0 km³ | 2022 | World Bank (A1) |
| Internal resources per inhabitant | 9,880 m³ | 2022 | World Bank (A1) |



*Rice harvest on a trial field at Waza (Far North).*

# 3. Sunshine, rainfall and climate change

## 3.1 Sunshine

Global solar radiation reaches 1,604 to 2,251 kWh/m² a year depending on the region. Energy is not the limit: Water available at the right time makes the harvest.

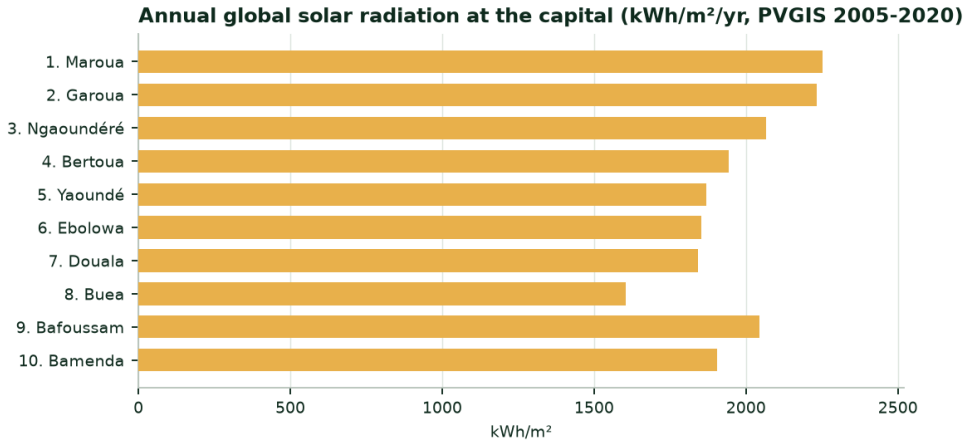



*Figure 3. Annual global solar radiation at each region's capital. Source: PVGIS (A8)*

## 3.2 Rainfall and evapotranspiration

It rains 4,057 mm a year at Buea, but only 569 mm at Maroua, where reference evapotranspiration reaches 2,240 mm. In the forested South, rain is spread over two seasons; in the North and the Far North, it falls in a single season of four to six months, followed by long dry months.

**Mean monthly rainfall, 1991-2020 (mm per month)**

| City | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Maroua | 0 | 0 | 0 | 9 | 36 | 67 | 124 | 192 | 112 | 29 | 1 | 0 |
| Garoua | 0 | 0 | 1 | 16 | 59 | 74 | 126 | 184 | 126 | 54 | 1 | 0 |

| City | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Ngaoundéré | 0 | 2 | 26 | 165 | 280 | 324 | 354 | 386 | 346 | 202 | 18 | 0 |
| Bertoua | 13 | 26 | 104 | 160 | 141 | 117 | 134 | 151 | 173 | 220 | 130 | 25 |
| Yaoundé | 40 | 57 | 128 | 172 | 178 | 161 | 133 | 129 | 190 | 248 | 174 | 54 |
| Ebolowa | 70 | 91 | 181 | 204 | 200 | 187 | 165 | 156 | 242 | 284 | 203 | 90 |
| Douala | 64 | 92 | 209 | 272 | 286 | 305 | 345 | 340 | 340 | 343 | 244 | 91 |
| Buea | 95 | 130 | 240 | 270 | 295 | 434 | 616 | 637 | 570 | 447 | 222 | 101 |
| Bafoussam | 2 | 8 | 41 | 115 | 145 | 155 | 238 | 263 | 233 | 179 | 47 | 3 |
| Bamenda | 7 | 26 | 124 | 274 | 290 | 324 | 351 | 340 | 379 | 294 | 78 | 7 |

*Source: ERA5 via Open-Meteo (A9); author's calculation.*

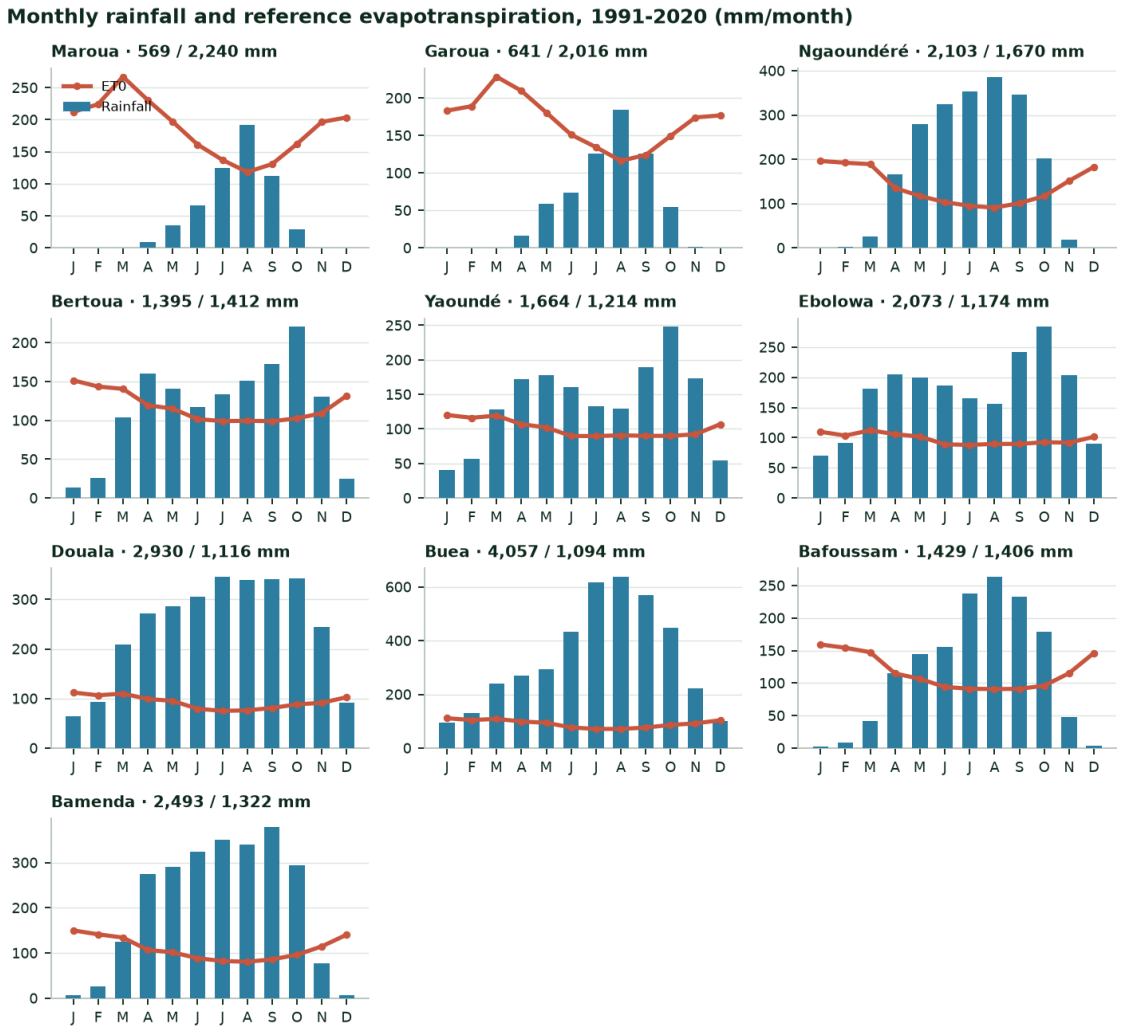



*Figure 4. Monthly rainfall and reference evapotranspiration by city, 1991-2020. Source: ERA5 via Open-Meteo (A9)*

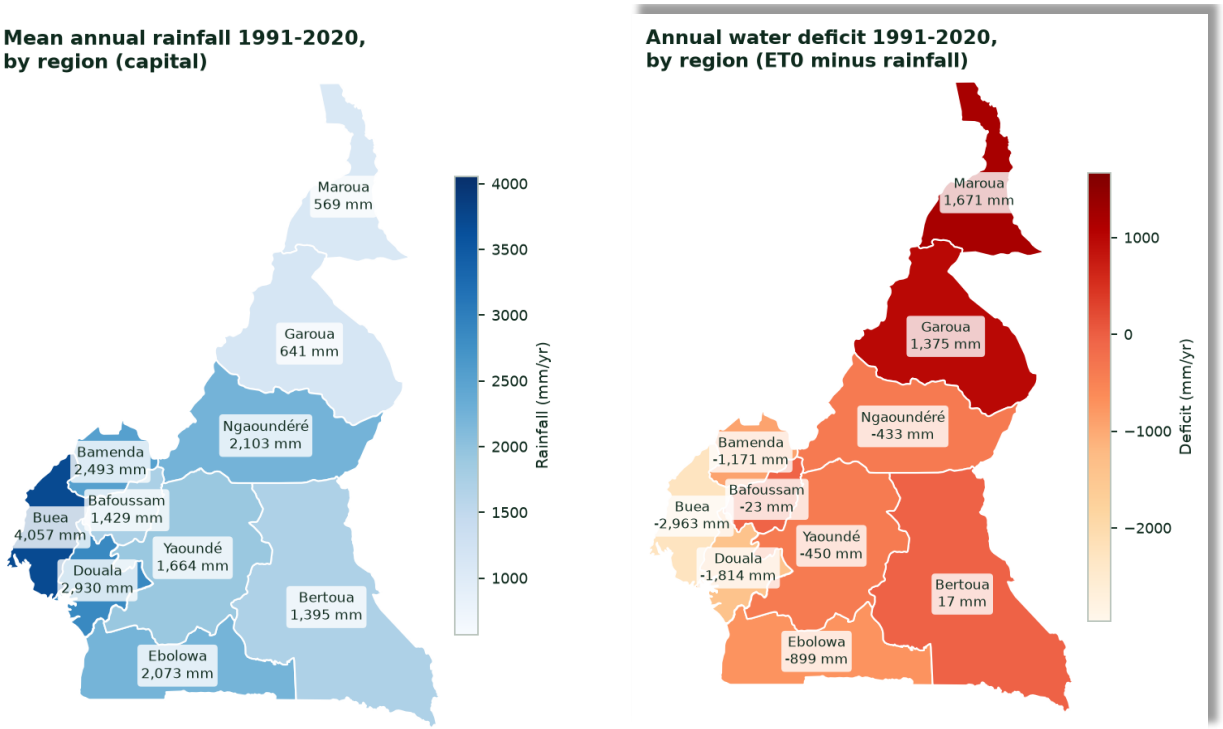



*Figure 5. Mean annual rainfall 1991-2020, by region. Source: ERA5 (A9), Natural Earth (A10). Annual water deficit, by region. Source: ERA5 (A9), Natural Earth (A10).*

## 3.3 Climate change

Mean annual temperature is rising. At Douala it went from 25.0 °C in 1981 to 26.7 °C in 2024, a trend of 0.26 °C per decade. At Maroua the trend is 0.14 °C per decade, from 27.1 °C to 27.9 °C. Warmer air dries the soil faster between two rains.

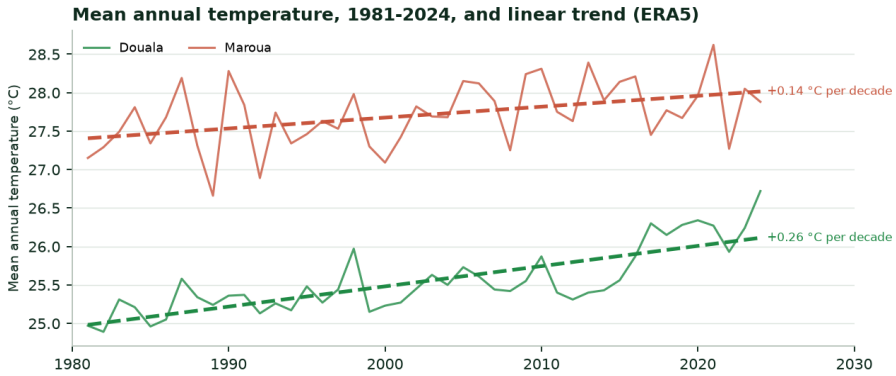



*Figure 6. Mean annual temperature, 1981-2024, and linear trend. Source: ERA5 (A9)*



*Rhumsiki and Kapsiki peak, in the Mandara mountains (Far North).*

# 4. Water: Resources, irrigated areas, withdrawals

## 4.1 Resources and withdrawals

Total freshwater withdrawals are 1,088 million m³ a year (2022). Agriculture withdraws 67.7% of it, that is 737 Mm³. Internal resources are 273.0 km³ a year and water stress is 1.6%: The country has water, but little of it is mobilised for crops.

**Freshwater: Withdrawals and resources**

| Indicator | Value | Year | Source |
|---|---|---|---|
| Total withdrawals | 1,088 Mm³ | 2022 | World Bank (A1) |
| Agricultural share | 67.7% | 2022 | World Bank (A1) |
| Agricultural withdrawals | 737.0 Mm³ | 2022 | Calculated from A1 |
| Area equipped for irrigation | 28,700 ha | 2024 | FAOSTAT (A6) |
| Agricultural withdrawals per equipped hectare | 25,679 m³/ha | — | Calculation (Annex B) |
| Water stress | 1.6% | 2022 | World Bank (A1) |

## 4.2 Irrigated areas

FAOSTAT counts 28,700 hectares equipped for irrigation in 2024. FAO estimates the country's irrigation potential at 290,000 ha (A11). Irrigation mostly serves rice in the large schemes of the Far North (SEMRY) and the Ndop plain, market gardening and sugar cane.



*Mangroves along the Atlantic, at the foot of Mount Cameroon.*

### 4.3 What the water saving represents

In the central scenario, the saving is 160.5 million m³ a year, or 21.8% of agricultural withdrawals. At 25,679 m³ per equipped hectare, this water would irrigate about 6,249 more hectares, or 21.8% of today's equipped area.

# 5. Fertilisers: Quantities, purchases, spending

## 5.1 Quantities used

In 2024, agricultural use reaches 52,002 tonnes of nitrogen, 13,782 tonnes of phosphate (P2O5) and 35,288 tonnes of potash (K2O). The country produces no nitrogen fertiliser: It imported 52,526 tonnes of nitrogen in the same year. Per hectare of arable land, use is 16.3 kg per hectare.

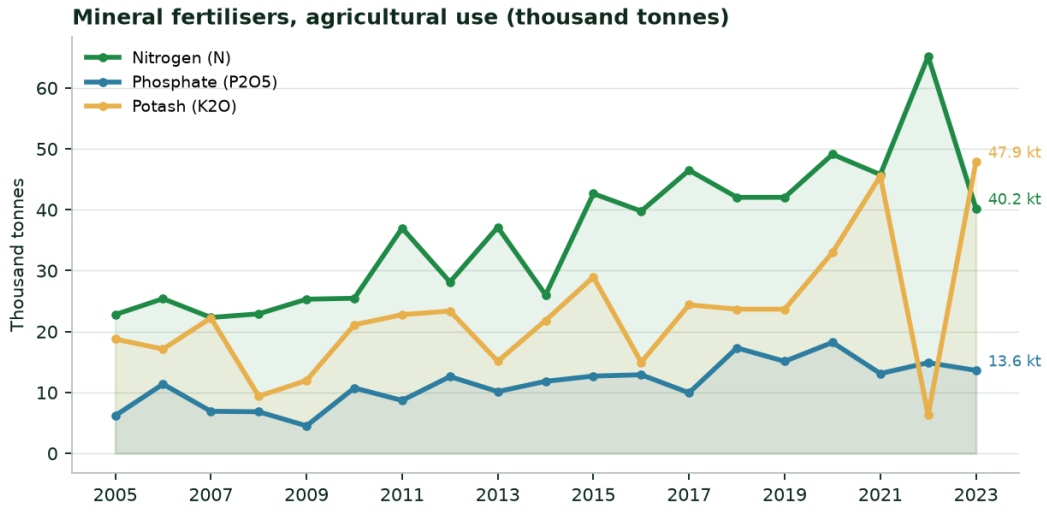



*Figure 7. Mineral fertilisers, agricultural use, 2005-2023. Source: FAOSTAT (A5)*

## 5.2 Regulatory framework

Law No. 2003/007 of 10 July 2003 governs the fertiliser sub-sector in Cameroon. A joint order of the Ministry of Agriculture (MINADER) makes fertilisers subject to a marketing authorisation before they are imported, produced or distributed (A13).



*Preparing the soil with compost, near Buea (South-West).*

## 5.3 Indicative spending

In 2024, the indicative minimum spending on mineral fertilisers reaches $70 million, at international fertiliser prices (calculation in Annex B, lower bound).

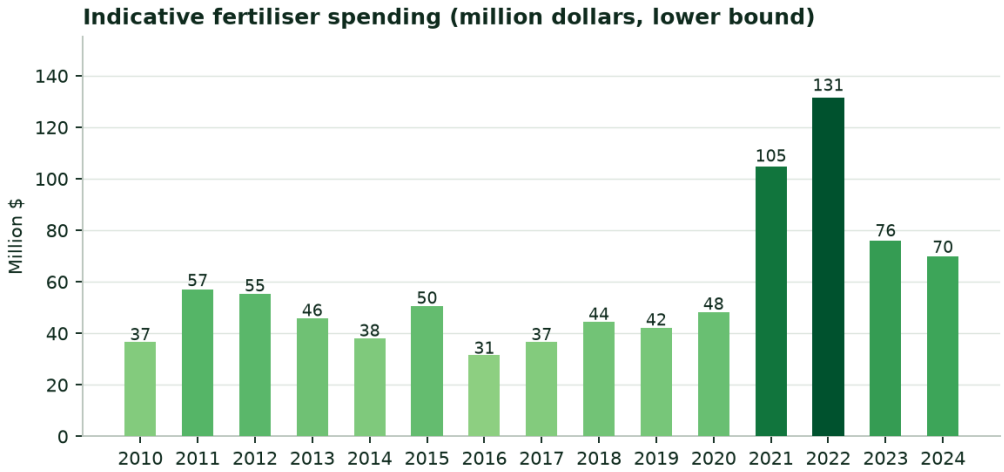



*Figure 8. Indicative fertiliser spending, lower bound. Source: Author's calculation from A5 and A7*

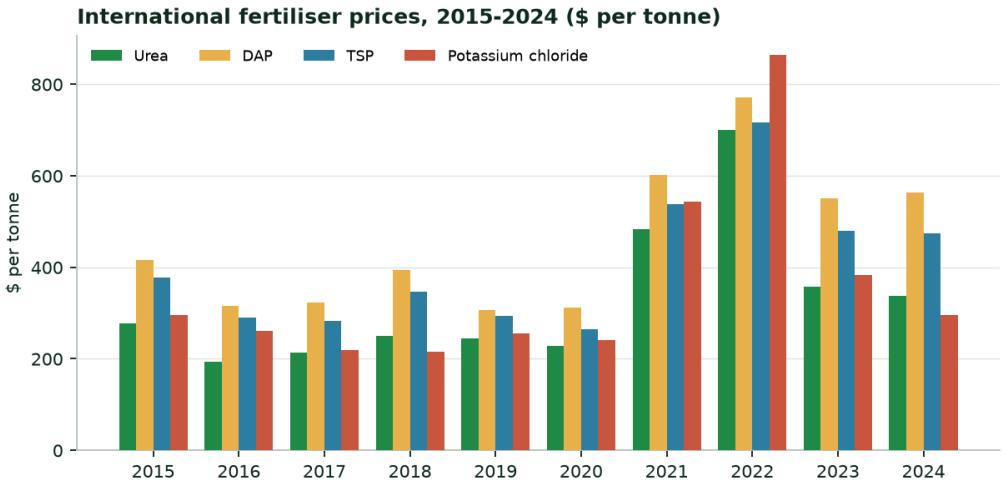



*Figure 9. International prices of urea, DAP, TSP and potassium chloride. Source: World Bank, Pink Sheet (A7)*

# 6. The regions at a glance

The table compares the country's ten regions using their capital. Climate figures are 1991-2020 averages, except radiation (2005-2020).

**Climate of the capitals, 1991-2020 means**

| Region (city) | Rainfall (mm/yr) | ET0 (mm/yr) | Deficit (mm/yr) | Radiation (kWh/m²/yr) | Temperature (°C) |
|---|---|---|---|---|---|
| Far North (Maroua) | 569 | 2,240 | 1,671 | 2,251 | 27.8 |
| North (Garoua) | 641 | 2,016 | 1,375 | 2,233 | 28.8 |
| Adamawa (Ngaoundéré) | 2,103 | 1,670 | -433 | 2,066 | 22.4 |
| East (Bertoua) | 1,395 | 1,412 | 17 | 1,942 | 24.4 |
| Centre (Yaoundé) | 1,664 | 1,214 | -450 | 1,869 | 23.1 |
| South (Ebolowa) | 2,073 | 1,174 | -899 | 1,852 | 23.2 |
| Littoral (Douala) | 2,930 | 1,116 | -1,814 | 1,842 | 25.6 |
| South-West (Buea) | 4,057 | 1,094 | -2,963 | 1,604 | 22.4 |
| West (Bafoussam) | 1,429 | 1,406 | -23 | 2,045 | 20.3 |
| North-West (Bamenda) | 2,493 | 1,322 | -1,171 | 1,905 | 20.7 |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

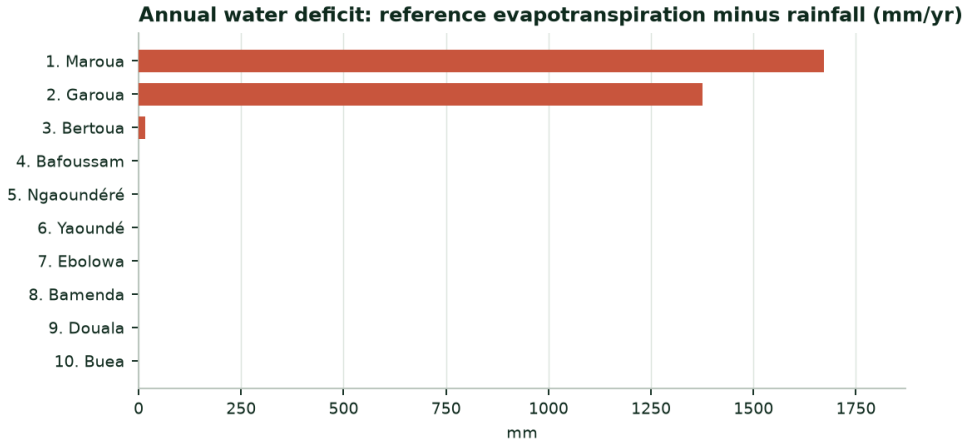



*Figure 10. Annual water deficit: Reference evapotranspiration minus rainfall. Source: ERA5 (A9)*



*Green hills in Cameroon.*

# 7. The twenty main crops: Three rankings

## 7.1 By harvested area

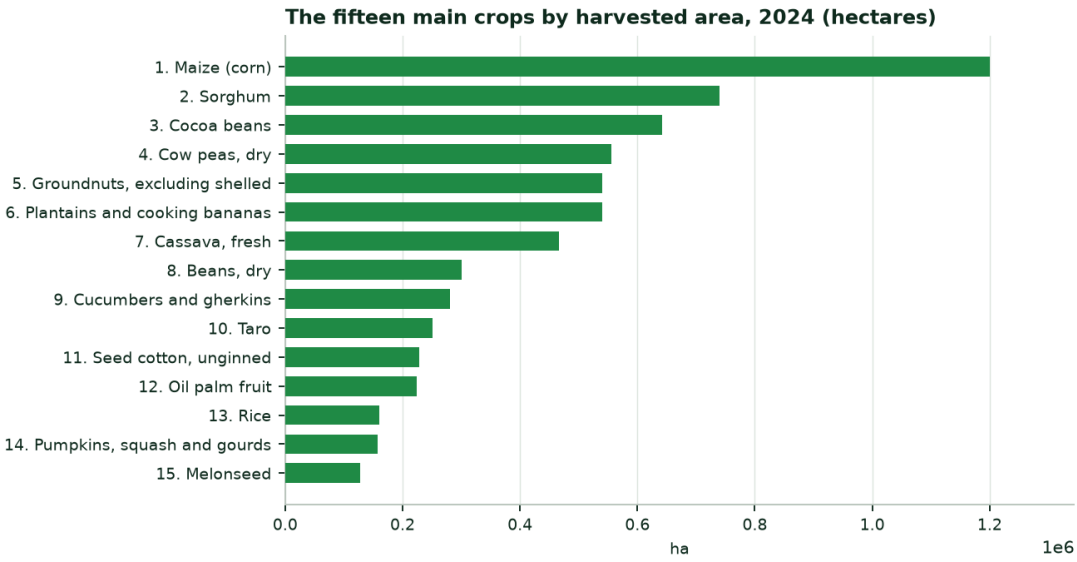



*Figure 11. The fifteen main crops by harvested area, 2024 (hectares). Source: FAOSTAT (A2)*

### The ten main crops by area, 2024

| Crop | Harvested area (ha) | Production (t) | Yield (t/ha) |
|---|---|---|---|
| Maize (corn) | 1,200,000 | 2,200,000 | 1.83 |
| Sorghum | 740,000 | 1,200,000 | 1.62 |
| Cocoa beans | 642,244 | 320,000 | 0.50 |
| Cow peas, dry | 555,235 | 250,312 | 0.45 |
| Groundnuts, excluding shelled | 540,000 | 540,000 | 1.00 |
| Plantains and cooking bananas | 539,375 | 4,663,350 | 8.65 |
| Cassava, fresh | 466,886 | 6,658,483 | 14.26 |
| Beans, dry | 300,202 | 388,217 | 1.29 |
| Cucumbers and gherkins | 280,269 | 265,462 | 0.95 |
| Taro | 250,535 | 1,946,970 | 7.77 |

*Source: FAOSTAT (A2).*

## 7.2 By production

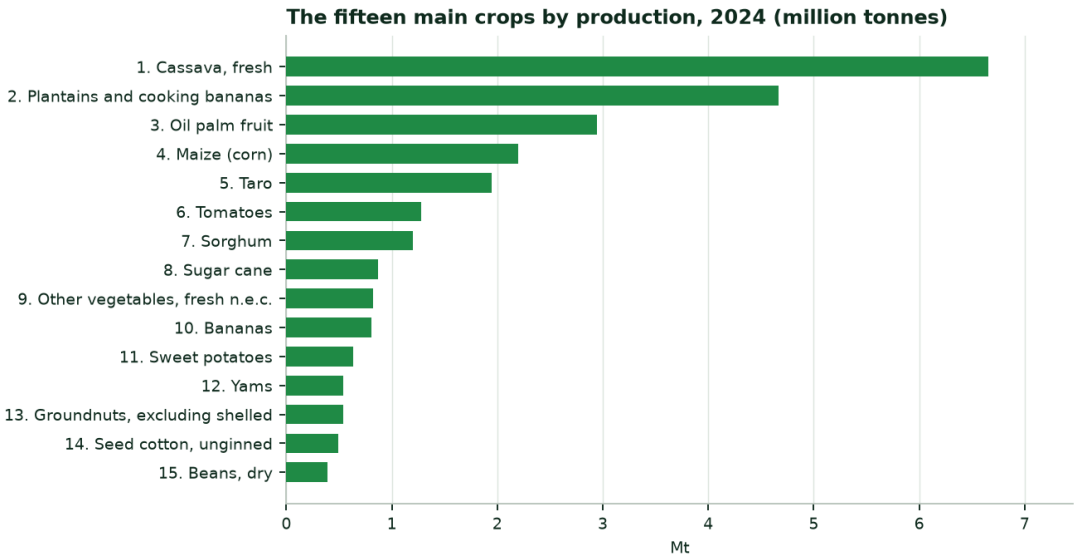



*Figure 12. The fifteen main crops by production, 2024 (million tonnes). Source: FAOSTAT (A2)*

## 7.3 By production value

By production value, the three main crops are: Plantains and cooking bananas, cassava, fresh and taro (FAOSTAT, 2023, constant international dollars).

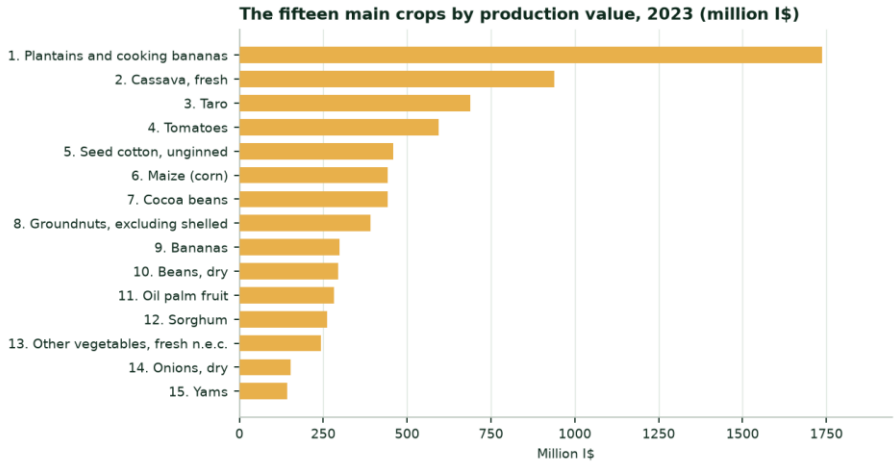



*Figure 13. The fifteen main crops by production value, 2023 (million I$). Source: FAOSTAT (A3)*



*A cocoa tree (Theobroma cacao).*

# 8. Extrapolation by crop: Irrigation and rainfed crops

## 8.1 Method

**Method in brief**

• **Irrigation.** National agricultural water is shared between irrigated crops (rice, vegetables and sugar cane), in proportion to their harvested areas; the equipped area is shared the same way.

• **Rainfed crops.** For crops that have a trial (maize, soybean and potato), the production gain is calculated per 10,000 ha treated, at the national average 2024 yield.

• Each calculation applies the effect of the trial cited: A quarter (cautious), half (central), all of it (high). Without a trial: 5, 10 and 20% of water, with no yield gain.

## 8.2 Trials used as analogues

**Trials cited and their use in the calculation**

| Trial | Crop | Water | Yield | Use in the calculation |
|---|---|---|---|---|
| Maize, Türkiye (TIGEM, Konya, 2025) | Maize | -38.9% | +17.1% | Maize: Yield |
| Soybean, China | Soybean | Assumption | +11.3% | Soybean: Yield |
| Watermelon, Mexico (2023) | Watermelon | -60.5% | +30.0% | Vegetables: Water and yield |
| Potato, South Africa (2025) | Potato | -52.0% | +31.4% | Potato: Yield |
| Date palm, UAE (ICBA, 2021) | Palm | -82.0% | +20.0% | Not used; cited for reference |

*Source: A14 to A18 (published trials and manufacturer's website, levels B and C).*

No trial has yet been conducted in Cameroon: The figures below apply the trials cited to the country's average yields.

## 8.3 Rainfed crops: Maize, Soybean and Potato

Average 2024 yields (FAOSTAT): Maize 1.83 t/ha, soybean 1.75 t/ha and potato 10.62 t/ha. A water-retention polymer keeps the water of each rain near the roots and helps the crop through dry days during the season.

### Extra production per 10,000 ha treated (tonnes a year)

| Crop | Average yield (t/ha) | Cautious | Central | High |
|---|---|---|---|---|
| Maize | 1.83 | 784 | 1,568 | 3,135 |
| Soybean | 1.75 | 495 | 991 | 1,981 |
| Potato | 10.62 | 8,334 | 16,669 | 33,338 |

*Source: FAOSTAT (A2); trials A14 to A18; author's calculation (Annex B).*

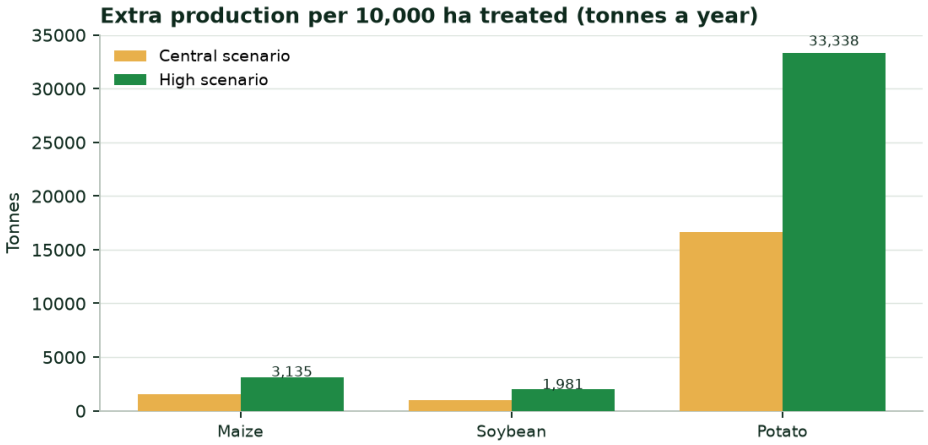



*Figure 14. Extra production per 10,000 ha treated, central and high scenarios. Source: Author's calculation*

## 8.4 Irrigation: Rice, Vegetables and Sugar cane

### Equipped area assigned, trial used and water saved (central scenario)

| Crop group | Equipped area (ha) | Share | Trial or assumption | Water saved (Mm³/yr) |
|---|---|---|---|---|
| Rice (paddy) | 11,040 | 38.5% | Generic assumption, no trial on rice | 28.4 |
| Vegetables (tomato, onion, okra, chilli, others) | 16,684 | 58.1% | Watermelon, Mexico (2023) | 129.6 |
| Sugar cane | 976 | 3.4% | Generic assumption, no trial | 2.5 |

*Source: Author's calculation (Annex B).*

### Water saved, extra irrigable hectares and production gain, by scenario

| Scenario | Water (Mm³/yr) | Extra hectares | Production (t/yr) |
|---|---|---|---|
| Cautious | 80.2 | 3,124 | 13,812 |
| Central | 160.5 | 6,249 | 27,624 |
| High | 320.9 | 12,497 | 55,249 |

*Source: Author's calculation (Annex B).*

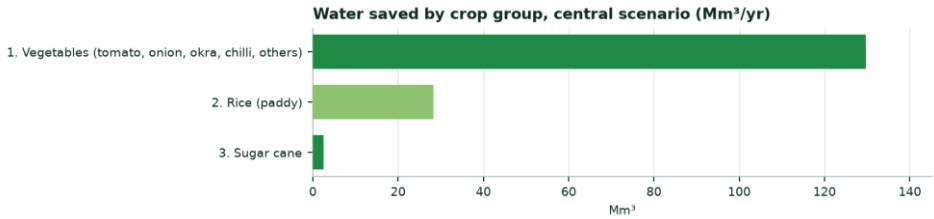



*Figure 15. Water saved by crop group, central scenario. Source: Author's calculation*

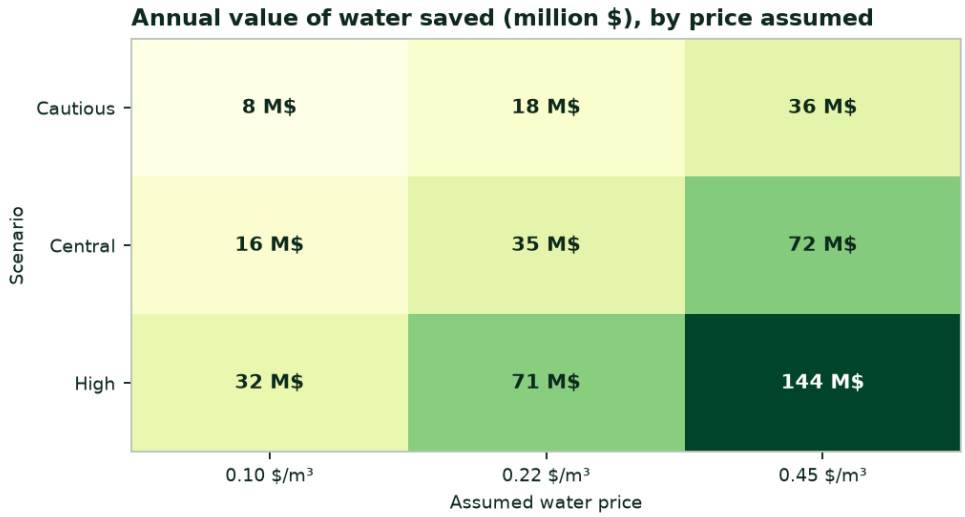



*Figure 16. Annual value of the water saved, by price assumed per cubic metre. Source: Author's calculation; prices assumed, no national tariff*

## 8.5 Reading the results

**Key points**

• In Cameroon, the main stake is rainfed farming: Only 28,700 ha are equipped for irrigation, for 7,967,020 ha of cropland.

• The "vegetables" group gives 81% of the irrigation water saving (watermelon, Mexico (2023)).

• The gains are counted by scenario and, in rainfed farming, per 10,000 ha treated: A cooperative or a public programme can scale them to its own area.

# 9. Potential benefits of our products

The table lists what the products may bring according to the documents available, with their confidence level and their limit.

**Potential benefits, by situation**

| Benefit | Published figures | Source (level) | Limit |
|---|---|---|---|
| Irrigation water saving | Date palm −82%; watermelon −60.5%; potato −52%; maize −38.9% | A14 to A18 (B, C) | One trial per crop; none in Cameroon |
| Yield | Maize +17.1%; potato +31.4%; watermelon +30%; date palm +20%; soybean +11.2 to +11.4% | A14 to A18 (B, C) | Trials conducted under irrigation or in other climates |
| Organic farming (ECOFERT®) | Certified by SOHISCERT and EU Bio | A21 (A) | Valid for the certified product |

| Benefit | Published figures | Source (level) | Limit |
|---|---|---|---|
| Safety: Acrylamide (ECOSORB®) | Below 3.0 µg/g | A19 (B) | One laboratory, one date |
| REACH compliance (EVERGREEN®) | ≤ 0.1% of substances of very high concern in 233 substances tested | A20 (B) | Screening, not an approval |
| Agroforestry (Paulownia ALTIFOLIA®) | 13 to 15 m in 6 to 8 years (website) | A22 (C) | No trial in Cameroon |

*Source: Annex A.*



*A banana plantation on a riverbank.*

# 10. External trade: Exports, imports, importing

## 10.1 Overview, 2023 and 2024

**Agricultural products and total merchandise trade (million $)**

| Indicator | 2023 | 2024 |
|---|---|---|
| Agricultural exports (crops and livestock) | 1,430 | 2,059 |
| Agricultural imports (crops and livestock) | 1,498 | 1,816 |
| Exports, all merchandise | 4,400 | 6,000 |
| Imports, all merchandise | 9,000 | 8,500 |

*Source: FAOSTAT (A4).*

## 10.2 Exports

The three main farm products exported in 2024 are: Cocoa beans ($1,170 million), cocoa butter, fat and oil ($296 million) and cocoa paste not defatted ($173 million).

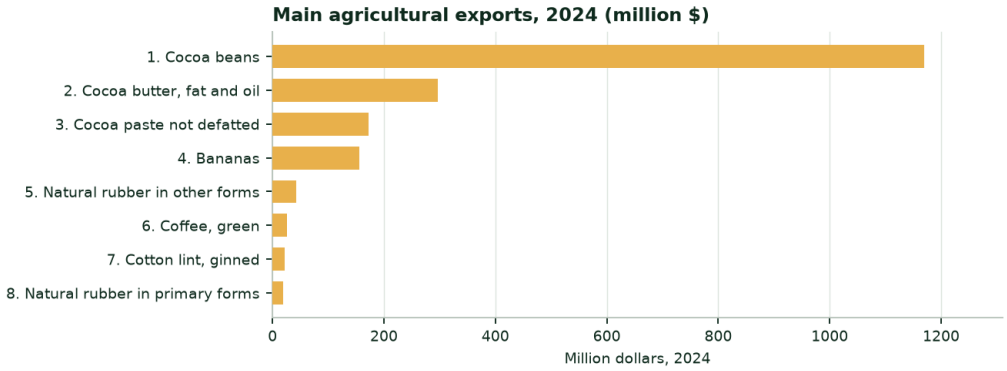



*Figure 17. Main agricultural exports, 2024. Source: FAOSTAT (A4)*

## 10.3 Imports

The main farm purchases are: Wheat ($288 million), refined sugar ($265 million) and malt, whether or not roasted ($81 million). Rice weighs $545 million (husked, broken and milled rice): Every tonne grown locally reduces this bill.

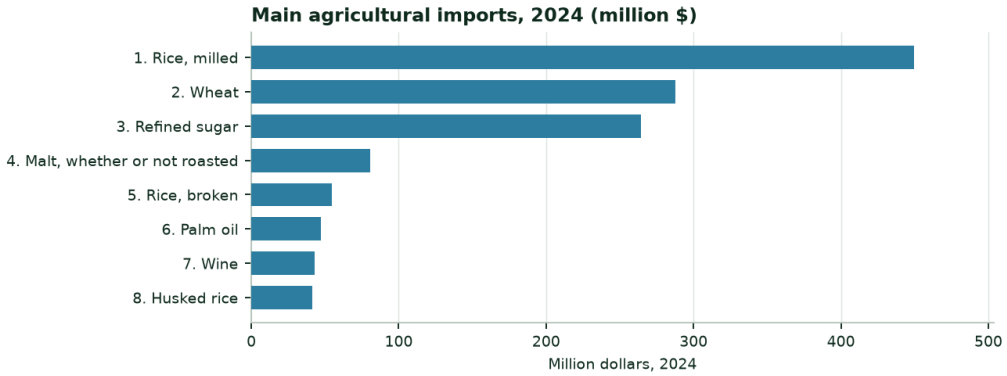



*Figure 18. Main agricultural imports, 2024. Source: FAOSTAT (A4)*



*Figure 19. Drying cocoa during the season.*



*Fresh vegetables at the market.*

## 10.4 Importing water-retention polymers into Cameroon

**What to know**

• **Customs duty (code 3906.90, acrylic polymers in primary forms):** ECCAS-CEMAC common external tariff, applied in Cameroon since 1 January 2026, with rates from 0 to 40% (A12). The rate of line 3906.90 is to be confirmed with the Directorate General of Customs.

• **Community levies:** 1% CEMAC community integration tax and 0.05% OHADA contribution (A12).

• **VAT:** 19.25%, that is 17.5% plus 10% of municipal surcharges (A12).

• **Approval and authorisation:** Importing and selling a fertiliser requires an approval and a marketing authorisation issued by the Ministry of Agriculture, MINADER (A13).

• **Classification:** Code 3906.90 is an inference for our products; Cameroonian customs settle it.

# 11. A study of each region

Each region is described through the city that represents it. The figures describe the climate; the water deficit (evapotranspiration minus rainfall) shows where water is most short. The description of crops is qualitative.



*Daybreak at Dschang, in the West region.*

## Far North (Maroua)

The driest region: Sorghum, millet, cotton, onion and livestock, with the irrigated rice of the SEMRY schemes along the Logone.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 569 mm |
| Reference evapotranspiration | 2,240 mm |
| Annual water deficit | 1,671 mm |
| Months with less than 50 mm of rain | 8 |
| Global solar radiation | 2,251 kWh/m² |
| Mean temperature | 27.8 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## North (Garoua)

Benue valley: Cotton, maize, sorghum and groundnuts; rain falls in a single season.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 641 mm |
| Reference evapotranspiration | 2,016 mm |
| Annual water deficit | 1,375 mm |
| Months with less than 50 mm of rain | 6 |
| Global solar radiation | 2,233 kWh/m² |
| Mean temperature | 28.8 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## Adamawa (Ngaoundéré)

A high plateau and cattle country: Maize, yam, cassava and potato.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 2,103 mm |
| Reference evapotranspiration | 1,670 mm |
| Annual water deficit | -433 mm |
| Months with less than 50 mm of rain | 5 |
| Global solar radiation | 2,066 kWh/m² |
| Mean temperature | 22.4 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## East (Bertoua)

A thinly populated forest region: Cassava, plantain, maize, cocoa and robusta coffee.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 1,395 mm |
| Reference evapotranspiration | 1,412 mm |
| Annual water deficit | 17 mm |
| Months with less than 50 mm of rain | 3 |
| Global solar radiation | 1,942 kWh/m² |
| Mean temperature | 24.4 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## Centre (Yaoundé)

With the South, the main cocoa basin: Cassava, plantain, groundnuts and market gardening around the capital.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 1,664 mm |
| Reference evapotranspiration | 1,214 mm |
| Annual water deficit | -450 mm |
| Months with less than 50 mm of rain | 1 |
| Global solar radiation | 1,869 kWh/m² |
| Mean temperature | 23.1 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## South (Ebolowa)

Dense forest: Cocoa, oil palm, rubber, cassava and plantain.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 2,073 mm |
| Reference evapotranspiration | 1,174 mm |
| Annual water deficit | -899 mm |
| Months with less than 50 mm of rain | 0 |
| Global solar radiation | 1,852 kWh/m² |
| Mean temperature | 23.2 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## Littoral (Douala)

Export bananas and oil palm in the Moungo, market gardening around Douala, under very heavy rainfall.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 2,930 mm |
| Reference evapotranspiration | 1,116 mm |
| Annual water deficit | -1,814 mm |
| Months with less than 50 mm of rain | 0 |
| Global solar radiation | 1,842 kWh/m² |
| Mean temperature | 25.6 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## South-West (Buea)

Volcanic soils of Mount Cameroon: Tea, bananas, oil palm, rubber, cocoa and coffee.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 4,057 mm |
| Reference evapotranspiration | 1,094 mm |
| Annual water deficit | -2,963 mm |
| Months with less than 50 mm of rain | 0 |
| Global solar radiation | 1,604 kWh/m² |
| Mean temperature | 22.4 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## West (Bafoussam)

Intensively farmed highlands: Maize, beans, potato, market gardening and arabica coffee.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 1,429 mm |
| Reference evapotranspiration | 1,406 mm |
| Annual water deficit | -23 mm |
| Months with less than 50 mm of rain | 5 |
| Global solar radiation | 2,045 kWh/m² |
| Mean temperature | 20.3 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## North-West (Bamenda)

North-West highlands: Maize, beans, potato, arabica coffee and rice from the Ndop plain.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 2,493 mm |
| Reference evapotranspiration | 1,322 mm |
| Annual water deficit | -1,171 mm |
| Months with less than 50 mm of rain | 3 |
| Global solar radiation | 1,905 kWh/m² |
| Mean temperature | 20.7 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

# Annex A. Sources and confidence levels

| Ref. | Source | Level |
|---|---|---|
| A1 | World Bank, World Development Indicators (WDI), Cameroon | A |
| A2 | FAOSTAT, crops and livestock (QCL), areas, production, 2014-2024 | A |
| A3 | FAOSTAT, value of production (QV), constant 2014-2016 I$, 2019-2024 | A |
| A4 | FAOSTAT, trade in crops and livestock products, 2018-2024 | A |
| A5 | FAOSTAT, mineral fertilisers, agricultural use, production and imports, 2005-2023 | A |
| A6 | FAOSTAT, land use, areas equipped for irrigation, organic farming | A |
| A7 | World Bank, Pink Sheet (world prices of urea, DAP, TSP and potassium chloride) | A |
| A8 | PVGIS (European Commission Joint Research Centre), radiation, 2005-2020 | B |
| A9 | ERA5 (Copernicus), via Open-Meteo: Rainfall, reference evapotranspiration, temperature, 1981-2024 | B |
| A10 | Natural Earth, first-level administrative boundaries and country outline | B |
| A11 | FAO, AQUASTAT, Cameroon country profile ("Irrigation in Africa in figures", 2005 survey): Irrigation potential | A |
| A12 | ECCAS-CEMAC common external tariff, applied in Cameroon since 1 January 2026 (Directorate General of Customs, reported by Agence Ecofin); CEMAC levies (TCI, OHADA); Cameroon general tax code (VAT) | A |
| A13 | Law No. 2003/007 of 10 July 2003 governing the fertiliser sub-sector; joint order No. 00115 of MINADER (marketing authorisation for fertilisers) | A |
| A14 | Soybean trial, China (yield), manufacturer's website | C |
| A15 | Date palm trial, UAE (ICBA, 2021), manufacturer's website | B |
| A16 | Watermelon trial, Mexico (2023), manufacturer's website | C |
| A17 | Potato trial, South Africa (2025), manufacturer's website | C |
| A18 | Maize trial, Türkiye (TIGEM, Konya, 2025), manufacturer's website | B |
| A19 | ECOSORB® acrylamide analysis (Polymex laboratory, 2024) | B |
| A20 | EVERGREEN® REACH / SVHC screening (SGS, 2023) | B |
| A21 | ECOFERT® certificate (SOHISCERT, EU Bio) | A |
| A22 | Paulownia ALTIFOLIA® sheet figures (website) | C |

# Annex B. Calculation method

**1. Agricultural water.** Total withdrawals (1,088 Mm³, 2022) × agricultural share (67.71%) = 737.0 Mm³.

**2. Irrigation.** The equipped area (28,700 ha) and agricultural water are shared between irrigated groups (rice, vegetables and sugar cane) in proportion to their harvested areas (415,929 ha), without exceeding each group's harvested area. Our sources have no irrigation statistic by crop. Other crops are treated as rainfed and receive no irrigation water in the calculation. Cocoa, coffee, bananas and oil palm are treated as rainfed.

**3. Scenarios.** Water saved = group water × trial effect: A quarter (cautious), half (central), all of it (high). Without a trial: 5, 10 and 20%.

**4. Extra hectares.** Water saved ÷ water per equipped hectare (25,679 m³/ha).

**5. Rainfed crops.** Gain per 10,000 ha = 10,000 × average 2024 yield (production ÷ harvested area, FAOSTAT) × trial effect on yield (maize 17.1%, soybean 11.3% and potato 31.4%), with the same quarter, half and whole.

**6. Value of water.** Three assumed prices (0.10; 0.22 and 0.45 $/m³); no national tariff was adopted.

**7. Fertilisers.** Minimum spending = nitrogen ÷ 0.46 × urea price + phosphate ÷ 0.46 × TSP price + potash ÷ 0.60 × potassium chloride price (A7).

**8. Climate.** Rainfall, reference evapotranspiration (FAO-56) and temperature: 1991-2020 means of the ERA5 grid cell containing each city; radiation: PVGIS, 2005-2020. Temperature trends are linear regressions over 1981-2024. ERA5 is a reanalysis: Its rainfall can differ from local station records. On the maps, each region takes the value of its capital.

**9. Trade.** Totals are those of the FAOSTAT crops and livestock line; products are listed one by one, without aggregates. Rice amounts add husked rice, broken rice and milled rice.

# Annex C. Photo credits

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*Mount Cameroon seen from Buea.*

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